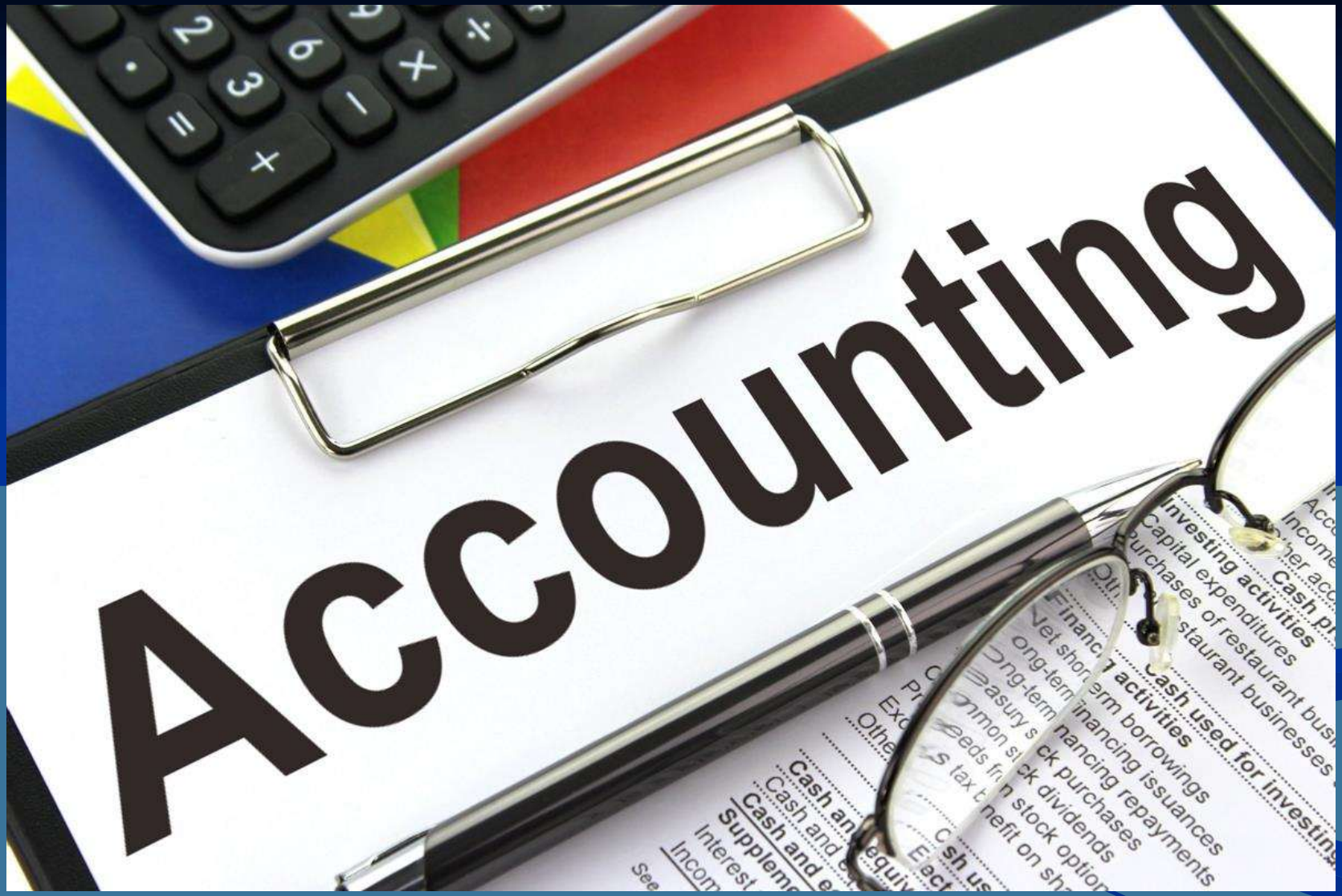


UNIT 3

Merchandise Inventory and Cost of Sales

Learning Objectives

1. Identify the components and costs included in merchandise inventory. (LO^1)
2. Calculate the cost of goods sold and merchandise inventory using specific identification, moving weighted average, and FIFO-perpetual. (LO^2)
3. Analyze the effects of the costing methods on financial reporting. (LO^3)

Learning Objectives

4. Calculate the lower of cost and net realizable value of inventory. (LO^4)
5. Analyze the effects of merchandise inventory errors on current and future financial statements-perpetual. (LO^5)
6. Apply both the gross profit and retail methods to estimate inventory. (LO^6)

Assigning Costs to Merchandise Inventory

Accounting for merchandise inventory requires several decisions, these include:

- Items included and their costs (should all items be included?).
- Costing Method. (specific identification, moving weighted average or FIFO)
- Merchandise Inventory System. (perpetual or periodic)
- Use of net realizable value or other estimates.

Understanding these decisions is the purpose of this unit.

LO 1

Items in Merchandise Inventory

Merchandise inventory includes all goods owned by a company and held for sale.

This seems straight forward, and it normally is, but there are a few instances where things can get a bit more confusing.

Goods in Transit

Goods on Consignment

Goods Damaged or Obsolete

Items in Merchandise Inventory

Goods in Transit

When items are in transit from seller to buyer, we need to figure out which business will include the items in their inventory.

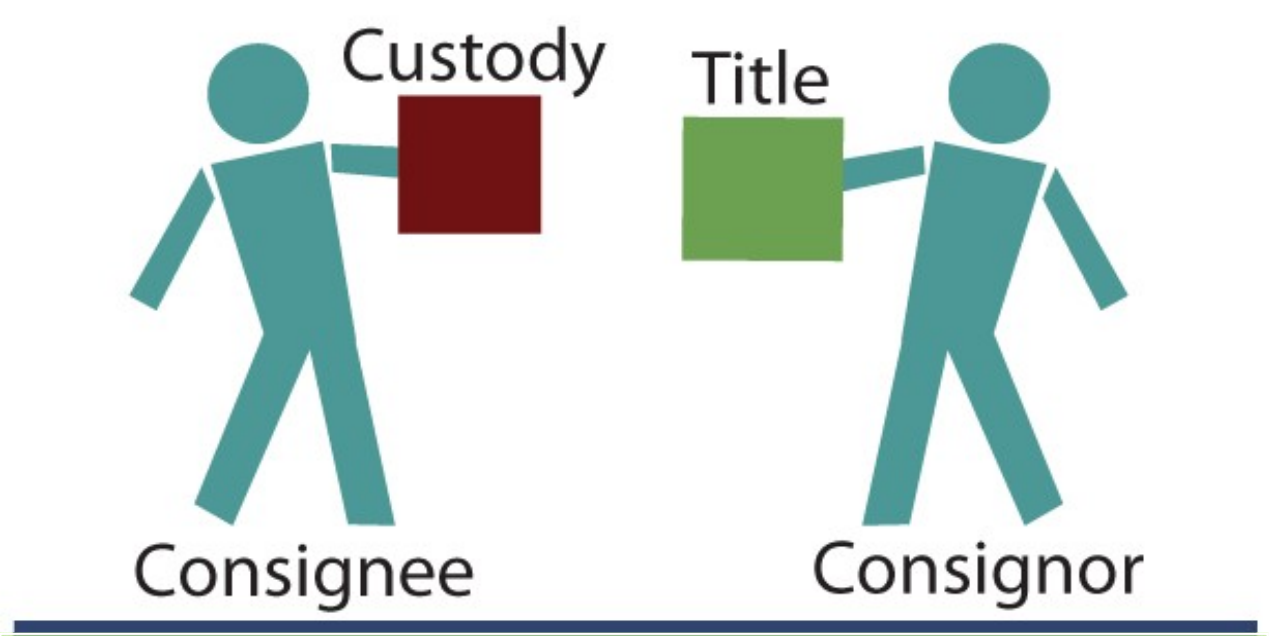
With F.O.B. Destination, the merchandise stays in the seller's inventory until delivery happens.

With F.O.B. Shipping Point, the merchandise moves to the buyer's inventory as soon as the merchandise leaves the factory.

Items in Merchandise Inventory

Goods on Consignment

These are goods shipped by their owner (***consignor***), to another party called (***consignee***). The consignee sells the product on behalf of the consignor but never takes legal title of the goods.



Consigned Goods Belong In the Inventory of the Consignor

Items in Merchandise Inventory

Goods Damaged or Obsolete

Damaged or obsolete (or deteriorated) goods are not counted in inventory <u>if they are unsaleable</u>.

If the goods are saleable at a reduced price, they are included in inventory at their ***net realizable value***.

Net realizable value (NRV) is sales price minus the cost of making the sale.

The period when damage, obsolescence, or deterioration occurs is the period where loss is reported.

Costs of Merchandise Inventory

Invoice price, less any discounts, plus all* expenditures necessary to bring an item to a saleable condition and location.

- This includes:
  - Import duties
  - Transportation-in
  - Storage
  - Insurance
  - Etc.

***The expenditures are always included in merchandise inventory as long as they are "material". Meaning that they have a meaningful impact on the value of the inventory.**

(GAAP – Materiality Principle)

States that an amount may not be ignored if its effect on the financial statements is important to their users.

Your Turn!

QS7-1 to QS 7-4

Physical Count of Merchandise Inventory

As we have discussed, a physical count of a merchandisers inventory is required periodically, even with perpetual systems, to ensure the accuracy of the records.

This will often occur at the end of a fiscal period, but can also happen at other points such as when inventory levels are low (makes the work easier).

Internal Controls

To ensure accuracy during this physical count, ***internal controls*** are followed to minimize errors.

Internal controls are policies and procedures used to protect assets, ensure reliable accounting, promote efficient operations, and urge adherence to company policies.

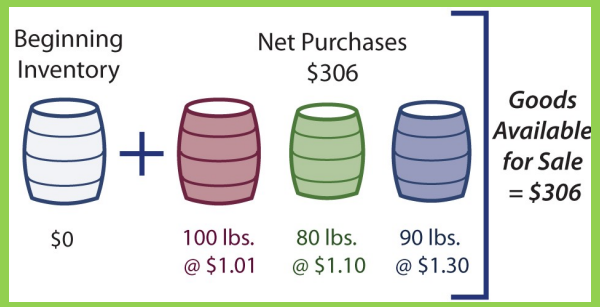
What are some internal controls for inventory?

A well-designed inventory control system should ensure:

- All purchases are authorized
- Accounting system ensures timely, accurate, and complete recording
- Receipt of inventory properly accounted for
- Inventory tested for quality when received/manufactured
- Costs properly identified and assigned to products
- Customer returns of inventory examined for defects
- Inventory reviewed for obsolescence
- New products introduced only after market studies and quality control tests have been made
- Management actively manages inventory
- Long term contracts are closely monitored

Assigning Costs to Merchandise Inventory

- **In a perfect world, all units in the inventory would be purchased at the same price. Unfortunately, this is rarely the case.**



Here you can see that three purchases were made, each of a different amount with a different cost.

- **Therefore, management must decide on a method of determining unit cost.**

- **This decision will affect both the income statement and the balance sheet accounts.**

- **Methods:**

1. **First-in, first-out (FIFO)**
2. **Moving weighted average**
3. **Specific identification**

LIFO…last in, first out.
- Popular in the U.S.
- Not allowed in Canada under GAAP or the Canadian *Income Tax Act*.

Your Turn!

Complete and turn in
MERCHADISE COST QUESTION

MERCHANDISE COST QUESTION - SOLUTION

OBC Corp has been selling a lot of Toronto Maple Leaf's hats as the hockey season is about to start. They are currently sold out.

On Monday Oct 1, the manager, Jill, drives to their supplier and buys 100 hats for cash at a cost of $15/hat. Journalize this purchase.

Date	Particulars	P.R.	Debit	Credit
Oct. 1	Merchandise Inventory		1,500	
	Cash			1,500
	Purchased 100 hats @ $15/hat			

MERCHANDISE COST QUESTION - SOLUTION

On Tuesday Oct. 2, OBC Corp. sold 70 hats for cash at a price of $50/hat. Journalize this transaction

Date	Particulars	P.R.	Debit	Credit
Oct. 2	Cash		3,500	
	Sales			3,500
	Sold 70 hats for $50/hat…cash			
2	Cost of Goods Sold		1,050	
	Merchandise Inventory			1,050
	Sold 70 hats…cost was $15/hat			

MERCHANDISE COST QUESTION - SOLUTION

On Wednesday Oct. 3, the manager, Jill, drives to their supplier and buys 60 hats for cash at a cost of $18/hat. The hats were identical to those purchased on Oct. 1, but the supplier has increased their price. Journalize this purchase.

Date	Particulars	P.R.	Debit	Credit
Oct. 1	Merchandise Inventory		1,080	
	Cash			1,080
	Purchased 60 hats @ $18/hat			

MERCHANDISE COST QUESTION - SOLUTION

The current inventory of hats would consist of:

				# of units on hand	M/I $Balance
Oct. 1:	Add:	100 x $15 =	$1,500	100	$1,500
Oct. 2:	Less:	70 x $15 =	$1,050	30	$450
Oct. 3:	Add:	60 x $18 =	$1,080	90	$1,530
Oct. 4:	Less	50 x $?? =	$?????	40	$?????

On Thursday Oct. 4, OBC Corp. sold 50 hats for cash at a price of $50/hat. Journalize this transaction

Date	Particulars	P.R.	Debit	Credit
Oct. 4	Cash		2,500	
	Sales			2,500
	Sold 50 hats for $50/hat…cash			
4	Cost of Goods Sold		????	
	Merchandise Inventory			????
	Sold 50 hats…cost was $??/hat			

Can we tell which hats were sold????

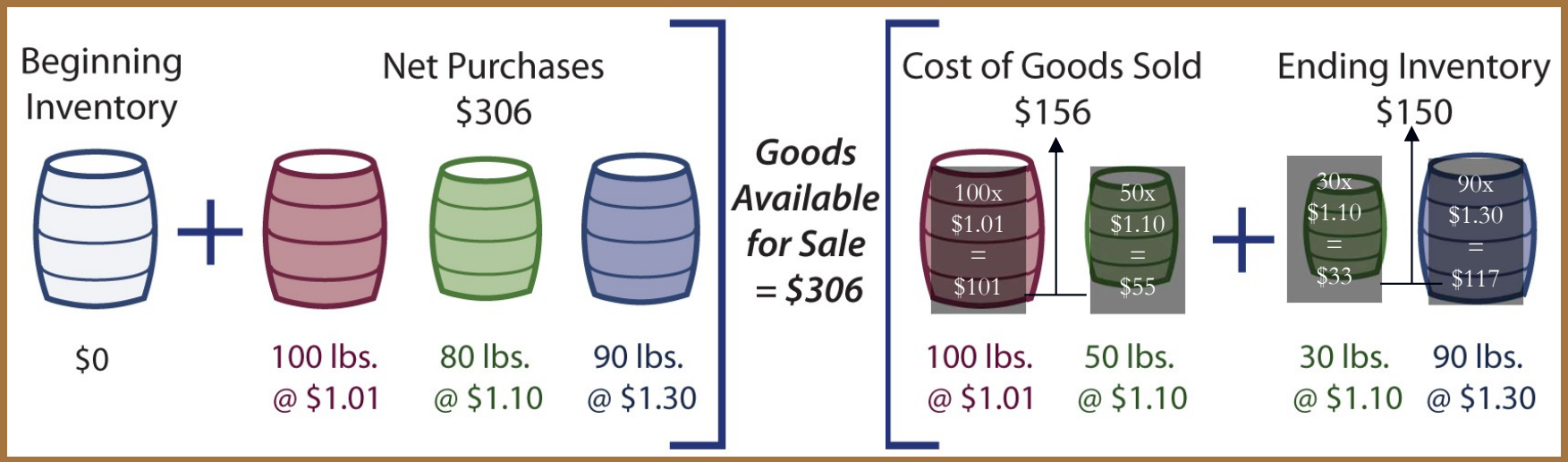
Do we sell the oldest hats first????

Can we keep track of our average cost per hat????

This is where an inventory costing method becomes important.

First-In, First-Out (FIFO)

- Based on the assumption that the items are <u>sold in the order acquired</u>.
- When a sale occurs:
  - The earliest units purchased are charged to Cost of Goods Sold.
  - The cost of the most recent purchases remain in merchandise inventory.



Lets assume we sell 150 pounds of product.

What is our:

- Cost of Goods Sold?
- Value of our ending inventory?

FIFO — Example

FIFO Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				10	$ 91	$ 910			
				9	$ 115	$ 1,035	11	$ 115	$ 1,265
Totals	**45**		**$ 4,800**	**34**		**$ 3,535**	**11**		**$ 1,265**

The opening inventory consists of 10 units @ $91/unit.

FIFO — Example

FIFO Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				10	$ 91	$ 910			
				10	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
8/28				5	$ 106	$ 530			
				9	$ 115	$ 1,035	11	$ 115	$ 1,265
Totals	**45**		**$ 4,800**	**34**		**$ 3,535**	**11**		**$ 1,265**

Additional units are purchased @ $106/unit.

This results in two layers of merchandise inventory.

FIFO — Example

FIFO Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				10	$ 91	$ 910			
				10	$ 106	$ 1,060	5	$ 106	$ 530
							5	$ 106	$ 530
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300

Under FIFO, units are assumed to be sold in the order acquired. Therefore, of the 20 units sold on August 14, the first 10 units come from beginning inventory. Therefore, those 10 units are removed from the inventory record based on the cost of those units of $91.

FIFO — Example

FIFO Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				10	$ 91	$ 910			
				10	$ 106	$ 1,060	5	$ 106	$ 530
							5	$ 106	$ 530
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300

The remaining 10 units sold on August 14th come from the next purchase, made on August 3rd. Therefore, these units are removed from the inventory record based on their cost of $106.

FIFO — Example

FIFO Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				10	$ 91	$ 910			
				10	$ 106	$ 1,060	5	$ 106	$ 530
							5	$ 106	$ 530
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300
8/28				5	$ 106	$ 530			
				9	$ 115	$ 1,035	11	$ 115	$ 1,265
Totals	**45**		**$ 4,800**	**34**		**$ 3,535**	**11**		**$ 1,265**

The ending inventory consists of the 5 remaining units from the August 3 purchase.

FIFO — Example

FIFO Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				10	$ 91	$ 910			
				10	$ 106	$ 1,060	5	$ 106	$ 530
							5	$ 106	$ 530
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300
8/28				5	$ 106	$ 530			
				9	$ 115	$ 1,035	11	$ 115	$ 1,265
Totals	**45**		**$ 4,800**	**34**		**$ 3,535**	**11**		**$ 1,265**

On Aug. 17th we purchased another 20 units @ $115/unit. Our inventory is back to having two layers.

FIFO — Example

FIFO Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				10	$ 91	$ 910			
				10	$ 106	$ 1,060	5	$ 106	$ 530
							5	$ 106	$ 530
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300
8/28				5	$ 106	$ 530			
				9	$ 115	$ 1,035	11	$ 115	$ 1,265
Totals	**45**		**$ 4,800**	**34**		**$ 3,535**	**11**		**$ 1,265**

On Aug 28th we sold 14 units. 5 of these would come from the $106 "layer" and the remaining 9 would come from the $115 "layer".

Remaining inventory is 11 units @ $115/unit

Mini-Quiz

A company that uses a perpetual merchandise inventory system made the following cash purchases and sales:

Jan. 1-Purchased 100 units at $10 per unit.

Feb. 5-Purchased 60 units at $12 per unit.

Mar.16-Sold for cash 140 units for $16 per unit.

a) Prepare journal entries to record the sale assuming a **FIFO** system is used.

Cash	**2,240**	
Sales (140x $16)		**2,240**
Cost of goods sold	**1,480**	
Merchandise Inventory (100x$10+40x12)		**1,480**

b) What is the value of their ending inventory?

Jan 1 – 100 @ $10/unit = $1,000
Feb 5 – 60 @ $12/unit = $720
$1,720

(100 @ $10/unit+ 40@$12/unit)

$1,720
- 1,480
$240

Or 20 units @ $12/unit = $240

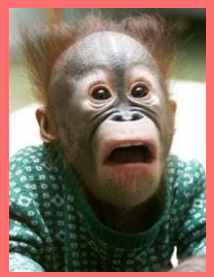
Your Turn!

- QS 7-5
- QS 7-6 part a only
- Ex 7-1 part a only
- Ex 7-3 1, 2, 3 part a only

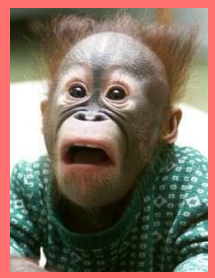
Moving Weighted Average Method

Under this method, the cost of all units are averaged together.

Purchase #1:	14 units @ $100/unit = $1,400 total cost	Total Inventory Value = $1,400 Average cost per unit = $100/unit ($1,400÷14)
Purchase #2:	25 units @ $110/unit = $2,750 total cost	Total Inventory Value = $4,150 ($1,400+$2,750) Average cost per unit = $106.41/unit ($4,150÷39)
Sale #1:	15 units @ $106.41/unit = $1,596.15 COGS	Total Inventory Value = $2,553.85 ($4,150-$1,596.15) Average cost per unit = $106.41/unit ($2,553.85÷24)
Sale #2:	22 units @ $106.41/unit = $2,341.02 COGS	Total Inventory Value = $212.83 ($2,553.85-$2,341.02) Average cost per unit = $106.42/unit ($212.83÷2)



While the majority of the time a sale does not change the cost per unit in the remaining inventory, there are times when rounding will impact the dollar value by a cent.



Moving Weighted Average - Example

Moving Weighted Average Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	(b) Units	(a)÷(b) Average Cost/Unit	(a) Total Cost
8-1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
8-3	15	$ 106	$ 1,590				25	$ 100	$ 2,500
									500
									2,800
8/28				14	$ 112	$ 1,568	11	$ 112	$ 1,232

The opening inventory consists of 10 units @ $91/unit.

Moving Weighted Average- Example

Notice that we no longer need to keep track of several "layers" of inventory.

Moving Weighted Average Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	(b) Units	(a)÷(b) Average Cost/Unit	(a) Total Cost
8-1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
8-3	15	$ 106	$ 1,590				25	$ 100	$ 2,500
									00
									00
8/28									32

Additional units are purchased @ $106/unit.

This results in an average cost of $100/unit.

$$\frac{(10 \times \$91) + (15 \times \$106)}{25 \text{ units}}$$

Moving Weighted Average- Example

Moving Weighted Average Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	(b) Units	(a)÷(b) Average Cost /Unit	(a) Total Cost
8-1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
8-3	15	$ 106	$ 1,590				25	$ 100	$ 2,500
8-14				20	$ 100	$ 2,000	5	$ 100	$ 500
							25	$ 112	$ 2,800
					$ 112	$ 1,568	11	$ 112	$ 1,232

These 20 units are sold at the average cost of $100/unit.

LO 2

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Moving Weighted Average- Example

Moving Weighted Average Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Total	(b) Units	(a)÷(b) Average Cost/Unit	(a) Total Cost
8-1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
8-3	15	$ 106	$ 1,590				25	$ 100	$ 2,500
8-14				20	$ 100	$ 2,000	5	$ 100	$ 500
8-17	20	$ 115	$ 2,300				25	$ 112	$ 2,800
8/28				14	$ 112	$ 1,568	11	$ 112	$ 1,232

An additional 20 units are purchased at a cost of $115/unit

This results in an average cost of $112/unit.

$$\frac{(5 \times \$100) + (20 \times \$115)}{25 \text{ units}}$$

LO 2

Moving Weighted Average- Example

Moving Weighted Average Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Total	(b) Units	(a)÷(b) Average Cost/Unit	(a) Total Cost
8-1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
8-3	15	$ 106	$ 1,590				25	$ 100	$ 2,500
8-14				20	$ 100	$ 2,000	5	$ 100	$ 500
8-17	20	$ 115	$ 2,300				25	$ 112	$ 2,800
8/28				14	$ 112	$ 1,568	11	$ 112	$ 1,232

14 items are sold at an average cost of $112/unit.

Means there are 11 items left in inventory at an average cost of $112.

LO 2

Mini-Quiz

A company that uses a perpetual merchandise inventory system made the following cash purchases and sales:

Jan. 1-Purchased 100 units at $10 per unit.

Feb. 5-Purchased 60 units at $12 per unit.

Mar.16-Sold for cash 40 units for $16 per unit.

Prepare journal entries to record the sale assuming a Moving Weighted Average system is used.

Date	Particulars	P.R.	Debit	Credit
Mar 16	Cash		640	
	Sales			640
	40 items sold at $16/unit			
16	Cost of Goods Sold		430	
	Merchandise Inventory			430
	((100x$10)+(60x$12))/160 = $10.75 x 40 units = $430			

Your Turn!

Day 1

- QS 7-8

Day 2

- QS 7-6 part b only
- Ex 7-1 part b only
- Ex 7-6

Specific Identification

This method is used when items:

- can be directly identified.
- can be directly identified with a specific purchase and its invoice. (typically when there is a relatively small number of items being sold.)

Examples: Automobiles, art, custom furniture.

Specific Identification — Example

Specific Identificaton Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
									182
									318
									182
							3	$ 106	$ 318
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300
8/28				2	$ 91	$ 182	3	$ 106	$ 318
				12	$ 115	$ 1,380	8	$ 115	$ 920
Totals	**45**		**$ 4,800**	**34**		**$ 3,562**	**11**		**$ 1,238**

The opening inventory consists of 10 units @ $91/unit.

Specific Identification — Example

Specific Identificaton Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
				8	$ 91	$ 728	2	$ 91	$ 182
									318
									182
									318
8/17	20	$ 115	$ 2,300						2,300
8/28				2	$ 91	$ 182	3	$ 106	$ 318
				12	$ 115	$ 1,380	8	$ 115	$ 920
Totals	**45**		**$ 4,800**	**34**		**$ 3,562**	**11**		**$ 1,238**

Additional units are purchased @ $106/unit.

This results in two layers of merchandise inventory.

Specific Identification — Example

Specific Identificaton Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				8	$ 91	$ 728	2	$ 91	$ 182
				12	$ 106	$ 1,272	3	$ 106	$ 318
							2	$ 91	$ 182
								$ 106	$ 318
								$ 115	$ 2,300
								$ 106	$ 318
								$ 115	$ 920
									$ 1,238

On August 14, 20 units are sold. Eight of these units came from the opening merchandise inventory and the remaining 12 units came from the August 3 purchase.

Specific Identification — Example

Specific Identificaton Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				8	$ 91	$ 728	2	$ 91	$ 182
				12	$ 106	$ 1,272	3	$ 106	$ 318
							2	$ 91	$ 182
							3	$ 106	$ 318
8							20	$ 115	$ 2,300
8							3	$ 106	$ 318
							8	$ 115	$ 920
Totals							**11**		**$ 1,238**

This leaves 2 units remaining from the original mercandise inventory and 3 units remaining from the August 3 purchase.

Specific Identification — Example

Specific Identificaton Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				8	$ 91	$ 728	2	$ 91	$ 182
				12	$ 106	$ 1,272	3	$ 106	$ 318
							2	$ 91	$ 182
							3	$ 106	$ 318
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300
8/28				2	$ 91	$ 182	3	$ 106	$ 318
				12	$ 115	$ 1,380	8	$ 115	$ 920
Totals	**45**		**$ 4,800**	**34**		**$ 3,562**	**11**		**$ 1,238**

20 more units are purchased Aug 17. This introduces a 3rd layer to the inventory.

Specific Identification — Example

Specific Identificaton Computations - Perpetual Merchandise Inventory System

	Purchases			Sales (at cost)			Inventory Balance		
Date	Units	Unit Cost	Total Cost	Units	Unit Cost	Cost of Goods Sold	Units	Unit Cost	Total Cost
8/1	Beginning Inventory								
	10	$ 91	$ 910				10	$ 91	$ 910
							10	$ 91	$ 910
8/3	15	$ 106	$ 1,590				15	$ 106	$ 1,590
8/14				8	$ 91	$ 728	2	$ 91	$ 182
				12	$ 106	$ 1,272	3	$ 106	$ 318
							2	$ 91	$ 182
							3	$ 106	$ 318
8/17	20	$ 115	$ 2,300				20	$ 115	$ 2,300
8/28				2	$ 91	$ 182	3	$ 106	$ 318
				12	$ 115	$ 1,380	8	$ 115	$ 920
Totals	**45**		**$ 4,800**	**34**		**$ 3,562**	**11**		**$ 1,238**

14 units are sold on Aug 17.
2 units from the $91 layer (1st layer) and
12 units from the $115 layer (3rd layer).

Your turn!

Answer these questions

QS 7-7

Ex 7-2

Ex 7-4

Complete This Question!

Problem 7-1A

Financial Reporting

- A company is required to use the same accounting methods from period to period (*consistency principle*).
- This does <u>not</u> mean that they must use one method exclusively.
  - For example, a business may use one costing method for one classification of inventory and a different for another.
- A change is only acceptable when it improves* financial reporting.
- If a change does take place, the business must add notes to the financial statements that report the type of change, its justification, and its effect on net income. (*full-disclosure principle*).

Does it Matter Which Inventory Costing Method We Use?

Results from Pr 7-1A

	Cost of Purchases	COGS	Inventory Balance
FIFO	**79,000**	**50,460**	**28,540**
Moving Weighted Average	79,000	50,387.40	28,612.60
Specific Identification	79,000	49,270	28,730

Financial Reporting

Advantages of Each Method

Moving Weighted Average

Smoothes out purchase price changes.

First-In, First-Out

Ending inventory approximates current replacement cost.

Specific Identification

Exactly matches costs and revenues.

Financial Reporting

Disadvantages of Each Method

Moving Weighted Average

Does not accurately match revenues to expenses.

First-In, First-Out

COGS does not reflect current costs. Oldest items at oldest prices are assigned to COGS.

Specific Identification

Relatively more costly to implement and maintain. Not feasible for many businesses.

Net Realizable Value

In 1995 BillyJack Enterprise purchased 3000 boxes of 3.5" computer disks at a cost of $15/box. They managed to sell most of these, but 500 boxes still remain in their merchandise inventory account at a value of $7,500.

What is the issue here?

Net Realizable Value

Net realizable value recognizes that items that are in merchandise inventory are usually worth more, but sometimes less, than what we paid for them.

So in other words, the value we could "realize" from the sale of items is usually more, but sometimes less, than the purchase price.

These discs are clearly worth less than the original purchase price.

Sometimes the value of our merchandise inventory increases for some reason.

Lower of Cost and Net Realizable Value (LCNRV)

- Merchandise Inventory is not always listed on the balance sheet at its actual cost.

States that we must exercise caution to ensure that assets and income are not overstated and that liabilities and expenses are not understated.

- The *prudence principle* requires that inventory be reported at net realizable value (NRV) **when NRV is lower than cost.**

- An adjusting entry is required if there is to be a reduction in value to reflect a lesser Net Realizable Value.

Lower of Cost and Net Realizable Value (LCNRV)

May be applied in one of two ways:

1. Separately to each item.
2. To groups of similar or related items.

Inventory Items	No. of Units	Cost/ Unit	Total Cost	NRV/ Unit	Total NRV	LCNRV applied to: Items	
Bicycles:							
Roadster	25	$ 750	$ 18,750				
Sprint	60	1,100	66,000				
Group subtotal							
Motorized Quads:							
A1 Series	21	$1,800	$ 37,800				
Trax-4	29	2,200	63,800				
Group subtotal							
Totals			$186,350				

Notice that the method used has an impact on the final values.

Journal Entries for Changes as a Result of LCNRV

Inventory Items	No. of Units	Cost/ Unit	Total Cost	NRV/ Unit	Total NRV	LCNRV applied to: Items	LCNRV applied to: Groups
Bicycles:							
Roadster	25	$ 750	$ 18,750	$ 790	$19,750	$ 18,750	
Sprint	60	1,100	66,000	1,100	66,000	66,000	
Group subtotal			$ 84,750		$85,750		$ 84,750
Motorized Quads:							
A1 Series	21	$1,800	$ 37,800	$1,300	$27,300	$ 27,300	
Trax-4	29	2,200	63,800	2,250	65,250	63,800	
Group subtotal			$101,600		$92,550		$ 92,550
Totals			$186,350			$175,850	$177,300

What would the adjusting entry look like in the journal in each case?

Item-by-item

Cost of Goods Sold	10,500	
Merchandise Inventory		10,500
To write inventory down to LCNRV;		
$186,350 – $175,850 = $10,500.		

Group method

Cost of Goods Sold	9,050	
Merchandise Inventory		9,050
To write inventory down to LCNRV;		
$186,350-$177,300 = $9,050		

Your Turn!

- QS 7-9
- QS 7-10
- Ex 7-7
- Problem 7-9A

Merchandise Inventory Errors

Errors in the computation, or physical count, of merchandise inventory will cause a <u>misstatement</u> of:

- Cost of goods sold
- Gross profit (Net Sales – Cost of Goods Sold)
- Net income
- Current assets
- Equity

Inventory Error Example

On Dec 31, 2017 a business has $930,000 in inventory in their warehouse. On the same date an employee tasked with counting the inventory reported, due to a counting error, that there was only $900,000 in inventory.

The accountant, assuming that the count was correct, would make the following adjusting journal entry.

2017		Dr	Cr
Dec 31	Cost of Goods Sold	$30,000	
	Merchandise Inventory		$30,000

This inventory counting error, and the resulting adjusting entry, would result in…

2017 Merchandise Inventory

930,000	30,000
900,000	

2017 Cost of Goods Sold

30,000	

	Item	Impact
Balance Sheet Items	Merchandise Inventory	Understated
	Current Assets	Understated
	Equity (Capital)	Understated
Income Statement Items	Cost of Goods Sold	Overstated
	Gross Profit	Understated
	Net Income	Understated

Inventory Error Example

At the end of the next year, Dec 31, 2018, the same employee went out to count the inventory. To keep things simple, let's assume that the business made no merchandise purchases or sales during the year.

If he counts correctly this time, what is the value of the inventory he will find?

$930,000

Since the ledger would now report $900,000, the accountant would need to make the following adjusting journal entry.

2018		Dr	Cr
Dec 31	Merchandise Inventory	$30,000	
	Cost of Goods Sold		$30,000

2018 Merchandise Inventory

900,000	
30,000	
930,000	

2018 Cost of Goods Sold

	30,000

Remember that Cost of Goods Sold is an expense, and would be closed at the end of each year. So, this years Cost of Goods Sold is being impacted by last year's mistake.

In the second year, the inventory error and the required adjusting entry for this year, is still having an impact…

Item	Impact
Merchandise Inventory	Now correct
Current Assets	Now correct
Equity (Capital)	Now correct
Cost of Goods Sold	Understated
Gross Profit	Overstated
Net Income	Overstated

While the balance sheet issues are corrected by the end of the second year, it is not until the third year that the income statement related errors are taken care of.

What Would the Effect be of a $2000 Overstatement in Ending Inventory?

	For years ended December 31, 2011, 2012, 2013, the income statement information should have been reported as:	Income statement information actually reported for years ended December 31, 2011	2012	2013
Sales	$30,000	$30,000	$30,000	$30,000
Cost of goods sold	20,000	18,000	22,000	20,000
Gross profit from sales	$10,000	$12,000	$ 8,000	$10,000
Operating expenses	5,000	5,000	5,000	5,000
Net income	$ 5,000	$ 7,000	$ 3,000	$ 5,000

2011 Adjusting Entry

M/I	2,000	
COGS		2,000

2012 Adjusting Entry

COGS	2,000	
M/I		2,000

Merchandise Inventory			
Jan. 1/11	10,000		
Purchases	20,000	18,000	COGS
Dec. 31/11	12,000		

Merchandise Inventory			
Jan. 1/12	12,000		
Purchases	20,000	22,000	COGS
Dec. 31/12	10,000		

Merchandise Inventory			
Jan. 1/13	10,000		
Purchases	20,000	20,000	COGS
Dec. 31/13	10,000		

1. 2011 net income and gross profit are overstated (too high) and cost of goods sold is understated (too low) when ending inventory is overstated (too high).
2. Ending inventory for one period becomes the beginning inventory for the next period, carrying forward any errors that existed.
3. 2012 net income and gross profit are understated (too low) and cost of goods sold is overstated (too high) when beginning inventory is overstated (too high). Notice that the error has reversed itself in 2012, the second year.
4. An inventory error in the year 2011 does not affect 2013.

Inventory Errors

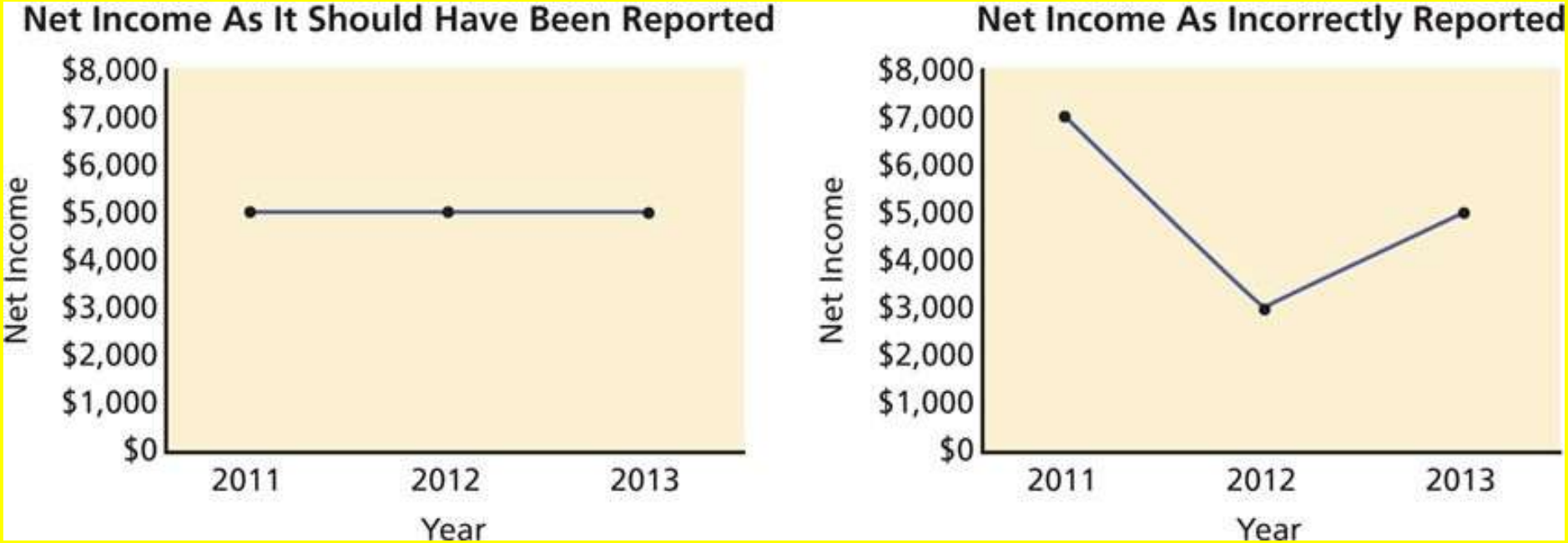
Effect on This Period's Income Statement

Inventory Error	Cost of Goods Sold	Net Income
Understate ending inventory	Overstated	Understated
Understate beginning inventory	Understated	Overstated
Overstate ending inventory	Understated	Overstated
Overstate beginning inventory	Overstated	Understated

"Self-Correcting" ≠ No Problem

Incorrect!!!

Imagine being a bank or potential investor seeing these two graphs. The graph on the right does not exactly instill confidence in the business.



Inventory Errors

Effect on This Period's Balance Sheet

Inventory Error	Assets	Equity
Understate ending inventory	Understated	Understated
Overstate ending inventory	Overstated	Overstated

A = L + O/E

Should be → 10 = 3 + 7

With Understatement → 8 = 3 + 5

With Overstatement → 12 = 3 + 9

Your Turn!

As a class we will complete the "Inventory Errors Example"

Your Turn!

- QS 7-11
- Ex 7-8
- Problem 7-7A <u>AND</u> 7-8A
- 1 non-textbook question